Weekly project: Employee

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_csv("Employee.csv")
print(df)
```

```
Age Attrition     BusinessTravel  DailyRate              Department  \
0     41       Yes      Travel_Rarely       1102                   Sales
1     49        No  Travel_Frequently        279  Research & Development
2     37       Yes      Travel_Rarely       1373  Research & Development
3     33        No  Travel_Frequently       1392  Research & Development
4     27        No      Travel_Rarely        591  Research & Development
...   ...       ...                ...        ...                     ...
1465   36        No  Travel_Frequently        884  Research & Development
1466   39        No      Travel_Rarely        613  Research & Development
1467   27        No      Travel_Rarely        155  Research & Development
1468   49        No  Travel_Frequently       1023                   Sales
1469   34        No      Travel_Rarely        628  Research & Development

      DistanceFromHome  Education EducationField  EmployeeCount  \
0                    1          2  Life Sciences              1
1                    8          1  Life Sciences              1
2                    2          2          Other              1
3                    3          4  Life Sciences              1
4                    2          1        Medical              1
...                ...        ...            ...            ...
1465                23          2        Medical              1
1466                 6          1        Medical              1
1467                 4          3  Life Sciences              1
1468                 2          3        Medical              1
1469                 8          3        Medical              1
```

	EmployeeNumber	...	RelationshipSatisfaction	StandardHours \
0	1	...	1	80
1	2	...	4	80
2	4	...	2	80
3	5	...	3	80
4	7	...	4	80
...	...	...	...	...
1465	2061	...	3	80
1466	2062	...	1	80
1467	2064	...	2	80
1468	2065	...	4	80
1469	2068	...	1	80

	StockOptionLevel	TotalWorkingYears	TrainingTimesLastYear \
0	0	8	0
1	1	10	3
2	0	7	3
3	0	8	3
4	1	6	3
...	...	...	...
1465	1	17	3
1466	1	9	5
1467	1	6	0
1468	0	17	3
1469	0	6	3

	WorkLifeBalance	YearsAtCompany	YearsInCurrentRole \
0	1	6	4
1	3	10	7
2	3	0	0
3	3	8	7
4	3	2	2
...	...	...	...
1465	3	5	2
1466	3	7	7
1467	3	6	2
1468	2	9	6
1469	4	4	3

```
     YearsSinceLastPromotion  YearsWithCurrManager
0                          0                     5
1                          1                     7
2                          0                     0
3                          3                     0
4                          2                     2
...                      ...                   ...
1465                       0                     3
1466                       1                     7
1467                       0                     3
1468                       0                     8
1469                       1                     2

[1470 rows x 35 columns]
```

```
df.head()
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	RelationshipSatisfac
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	...	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	...	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	...	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	...	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	...	

5 rows × 35 columns

```
df["Attrition"] = df["Attrition"].map({"Yes":1, "No":0})
df = pd.get_dummies(df, drop_first=True)


X = df.drop("Attrition", axis=1)
y = df["Attrition"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LogisticRegression(max_iter=20000,solver='liblinear')
model.fit(X_train, y_train)
```

LogisticRegression

▼ Parameters

penalty	'12'
dual	False
tol	0.0001
C	1.0
fit_intercept	True
intercept_scaling	1
class_weight	None
random_state	None
solver	'liblinear'
max_iter	20000
multi_class	'deprecated'
verbose	0
warm_start	False
n_jobs	None
l1_ratio	None

```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(accuracy)
0.8945578231292517
```